

The steepest descent direction is the direction opposite to the gradient. We have

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} x_1 \\ 4x_2 \end{pmatrix}.$$

At point $\mathbf{x} = (1 \ 1)^T$, we have

$$\nabla f(\mathbf{x}) = \begin{pmatrix} x_1 \\ 4x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Therefore, the steepest descent direction is

$$\begin{pmatrix} -1 \\ -4 \end{pmatrix}.$$

The directional derivative is

$$-\nabla f(\mathbf{x})^T \nabla f(\mathbf{x}) = (-1 \ -4) \begin{pmatrix} 1 \\ 4 \end{pmatrix} = -17.$$